KRISHISETU

Agricultural Market Discovery Platform

Project idea • Prototype implementation • System architecture • Features • Impact

CORE VALUE Help farmers understand crop-price benchmarks, find suitable buyers, compare selling options and estimate net earnings through market data, location-based discovery and regional-language voice assistance.

Document purpose

This document consolidates the KrishiSetu project idea and the implementation described in the supplied prototype overview. It is intended as a project attachment for a hackathon submission, demo, review or presentation. It distinguishes implemented prototype capabilities from configuration-dependent integrations and future enhancements.

Project at a glance

Project	KrishiSetu
Category	Agricultural technology / market discovery / voice-enabled assistance
Primary users	Farmers, agricultural buyers, wholesalers, APMC/market users
Core challenge	Fragmented price information, limited buyer discovery and unclear net earnings
Prototype frontend	Next.js, React, TypeScript
Backend and data	Express.js REST API, MongoDB Atlas
Voice stack	Gnani Prisma v2.5 speech-to-text and Timbre v2.5 text-to-speech
Primary documented coverage	16,307 price records, 10 APMC contact records, 41 mandi locations, 200+ commodities

1. Executive summary

KrishiSetu is a farmer-focused agricultural market discovery platform designed to make crop-market information easier to access and act on. It brings together government market-price observations, market and APMC contact information, registered buyer discovery, location-based maps, regional-language voice interaction, weather information and a net-return calculator in one application.

A farmer can ask a crop-price question in Kannada or English, search for markets or buyers, compare an available market benchmark with a buyer offer, estimate take-home earnings after selected costs, and contact a relevant market or buyer. The prototype uses a multi-source architecture and labels data with source and freshness information where available, so the user can distinguish published benchmarks from buyer-reported offers.

IMPORTANT SCOPE NOTE The prototype overview documents a functional demo with real data retrieval and connected services. It also identifies limits: the prediction engine is a simple baseline, APMC contact coverage is still limited, and some optional integrations require API keys.

2. Problem statement

Farmers may have difficulty finding a reliable reference price for their crop, knowing whether a local offer is competitive, locating alternative buyers, and estimating the amount they will receive after transportation and handling costs. Relevant information can be distributed across government price sources, market directories, buyer contacts and weather services. Language and digital-access barriers can make these tasks harder.

Problems KrishiSetu addresses

- Limited visibility into reported crop prices across mandis and nearby markets.
- Dependence on a single local buyer when alternatives are difficult to discover.
- Difficulty comparing a buyer's quoted price with a market benchmark.
- Uncertainty about net earnings after transport and handling deductions.
- Difficulty finding useful APMC contact details and market locations.
- Need for a more accessible way to ask agricultural-market questions in a regional language.

3. Proposed solution

KrishiSetu combines market-price intelligence, a buyer and market directory, a map-based discovery experience, a Kannada/English voice assistant and a net-return calculator. Rather than relying on one source or one headline price, it helps users inspect available source information, compare options and contact markets or buyers directly.

Core value proposition

KNOW • COMPARE • CONNECT Know the available market benchmark. Compare market and buyer options. Estimate net earnings. Connect with a relevant buyer or market contact.

Primary users

- Farmers checking reported prices and looking for buyers for a crop.
- Farmers comparing a vendor quote against an available market benchmark.
- Farmers estimating net proceeds for a given quantity and selling location.
- Buyers and registered businesses publishing crops purchased and buying requirements.
- Users looking for APMC/market contacts, locations and directions.

4. Main features and prototype implementation

4.1 Voice Agriculture Agent

The prototype integrates Gnani Prisma v2.5 for speech-to-text and Gnani Timbre v2.5 for text-to-speech. It supports Kannada and English interaction, uses an intent router with 14 intent types, retrieves data from MongoDB rather than relying on mock answers, and can speak responses back to the user.

- Example: ask for the reported tomato price in a mandi.
- Example: ask for buyers for a crop in a district.
- Example: request an earnings calculation for a stated quantity.

4.2 Multi-source market-price data

The application uses a multi-source architecture for market-price and directory data. The documented adapters include data.gov.in, AGMARKNET/e-NAM CSV imports, an APMC directory, Google Places discovery (API-key dependent), and the internal KrishiSetu buyer directory.

- Price observations include crop/commodity, market, reporting date, source and price fields where available.
- The documented prototype database contains 16,307 price records from data.gov.in.
- The overview describes CSV import for official AGMARKNET/e-NAM dashboard exports; it does not establish that every source is connected as a live streaming feed.

4.3 Market and APMC contact directory

The directory presents market name, market type, state/district/taluk, address and available public contact details. Geocoded locations and verification status support discovery. Contact actions include call, email, website and directions where those details exist.

- The supplied prototype overview reports 10 APMC contact records.
- Contact details and verification status should be interpreted according to the displayed source and status.
- The example APMC record in the source file is illustrative of the documented schema.

4.4 Interactive market map

A Leaflet map shows mandi locations, buyer markers and the user's location when location permission is granted. Marker popups can show names, districts, prices and contact information where available.

- The overview reports 41 mandi locations across India.
- The documented visual legend distinguishes mandi locations, verified buyers, pending buyers and the user location.
- Map tiles use OpenStreetMap and/or MapTiler; MapTiler configuration is optional with fallback described in the prototype notes.

4.5 Buyer discovery and contact

Farmers can search registered buyers and businesses, use geospatial search, view buyer details and contact options, and export buyer/farmer contact details as CSV. The documented data model supports buyer category, district, crops purchased, verification status and buyer offers.

- MongoDB 2dsphere indexing is documented for distance-based queries.
- The prototype overview describes statewide Karnataka discovery by default and configurable service areas.
- The buyer directory's usefulness depends on the number, freshness and verification of participating buyer records.

4.6 Price comparison and net-return calculator

The calculator compares an available mandi benchmark with buyer offers and estimates take-home earnings after configured costs. It supports quantity in kilograms, quintals and tonnes, with transport and handling costs included in the documented implementation.

- The prototype notes a fixed handling deduction of ₹1.20/kg.
- The displayed example calculates 500 kg at ₹20/kg with zero transport and ₹1.20/kg handling as ₹9,400 net return.
- Actual deductions and final proceeds can vary; users should verify transport, commissions, quality deductions, pickup and payment terms with the buyer.

4.7 Crop-price prediction

The documented prediction engine uses historical MongoDB price records and a baseline-last-observed method. It checks whether there are enough observations before generating a forecast and can report MAE/RMSE when backtesting is possible.

- The prototype notes a minimum of 10 observations before forecasting.
- This is a simple baseline, not a guarantee of future market prices.
- The source notes limitations including no seasonality in the baseline and the need for more historical data.

4.8 Weather integration

The prototype integrates Open-Meteo for current weather conditions and a seven-day forecast, including temperature, humidity, wind and precipitation fields where available. District-specific coordinates are documented for Karnataka districts.

- Weather is contextual information and should not be presented as a crop-price guarantee.
- Forecast coverage and accuracy depend on the selected location and external service.

4.9 Data status and source transparency

The architecture includes source health/status endpoints and source attribution. The project emphasizes transparent data sourcing and clear limitations rather than fabricating missing prices or predictions.

- The application should distinguish published market benchmarks, buyer-reported prices, farmer-reported quotes, historical observations and confirmed current offers.
- A periodic check of a source does not mean the source itself publishes new prices at that frequency.

5. End-to-end user journey

1. **Ask or search —** The farmer chooses a crop and location or asks a question by voice in Kannada/English.
2. **Understand intent —** The voice service transcribes speech and the intent router identifies the request type.
3. **Retrieve available information —** The backend queries stored market-price observations, market records, registered businesses and relevant service integrations.
4. **Compare options —** The interface shows the available benchmark, buyer information, location and source/date details.
5. **Estimate net earnings —** The user enters quantity and applicable costs to estimate the amount remaining after selected deductions.
6. **Connect and verify —** The farmer uses the contact/enquiry flow to speak with a buyer or market and confirms price, quantity, pickup and payment terms.

6. System architecture

The prototype follows a web frontend + REST backend + database architecture, with external service adapters for market data, voice, maps and weather.

Frontend	Next.js App Router, React, TypeScript, custom CSS variables, Lucide React
Map and location	Leaflet, OpenStreetMap/MapTiler, geocoded market/buyer coordinates
Backend	Express.js 5, TypeScript, REST API under /api/v1
Database	MongoDB Atlas with Mongoose; market prices, markets, businesses, buyer offers, voice interactions and predictions
Speech services	Gnani Prisma v2.5 STT; Gnani Timbre v2.5 TTS
Market data	data.gov.in API; AGMARKNET/e-NAM CSV adapter; APMC directory; internal buyer directory; Google Places adapter when

	configured
Weather	Open-Meteo API
Validation and utilities	Zod, Axios, Multer, WAV audio recording/encoding

7. Key backend modules and API areas

The supplied prototype overview documents these API groups and representative routes:

- Voice: POST /api/v1/voice/transcribe, POST /api/v1/voice/speak, POST /api/v1/voice/assistant, GET /api/v1/voice/status.
- Market prices: GET /api/v1/prices, GET /api/v1/prices/status, GET /api/v1/prices/filters, GET /api/v1/prices/fair-price.
- Market directory: GET /api/v1/markets, GET /api/v1/markets/:id, GET /api/v1/markets/:id/contact, GET /api/v1/markets/map.
- Predictions: POST /api/v1/predictions/crop-price, GET /api/v1/predictions/history.
- Buyers/businesses: GET /api/v1/buyers/search, GET /api/v1/buyers/nearby, GET /api/v1/businesses/me, POST /api/v1/enquiries.
- Admin/data operations: POST /api/v1/admin/sources/sync, POST /api/v1/admin/markets/import, POST /api/v1/admin/prices/import-csv, GET /api/v1/admin/health.

8. Data models

MarketPrice	Price observations, including state, district, market, commodity, normalized price where available, arrival/reporting date and source.
Market	Market/APMC directory record, market type, district, public phone/email and coordinates.
Business	Registered business/buyer profile, category, district, crops purchased and verification status.
BuyerOffer	Buyer-linked offer with crop, price, unit, quantity requirements and expiry.
CropPrediction	Stored price forecast information and related metadata.
VoiceInteraction	Voice request/response history and interaction data.

9. Prototype data and validation notes

16,307	Price records reported from data.gov.in
10	APMC contact records
41	Mandi locations with coordinates
200+	Commodity/crop types listed in the prototype overview
5	Documented source adapters
14	Intent types in the voice intent router
25+	REST endpoints documented in the development metrics
20+	React components documented in the development metrics

The prototype notes successful MongoDB Atlas connectivity, successful sample data.gov.in API retrieval, MapTiler geocoding for Davanagere, working Gnani STT authentication, generated Gnani TTS output, OpenRouter model response and a passing production build. These are the validation items stated in the supplied overview; they are not a substitute for a new independent test run.

10. Implementation status

Documented as completed

- Multi-source architecture with five adapters.
- Intent router with 14 intent types and MongoDB-backed data retrieval.
- Market contact directory and buyer/business search.

- Baseline price-prediction service and Open-Meteo weather service.
- Statewide price discovery, source-health dashboard and source attribution.
- Gnani-powered voice agent using STT and TTS.
- Interactive map with 41 mandi locations.
- Buyer/contact CSV export.
- Production build passes according to the supplied overview.

Pending or configuration-dependent

- Agricultural guidance for sowing/harvest recommendations is listed as pending.
- Comprehensive geocoding for all markets is pending.
- Google Places live testing requires an API key.
- More APMC directory/contact data is needed for broader production coverage.
- Longer historical price series are needed for stronger prediction performance.
- Real-time alerts, mobile app, farmer profiles, WhatsApp/SMS and QR scanning are future enhancements in the prototype notes.

11. Expected impact

- Improves access to available crop-price observations and market benchmarks.
- Makes it easier to discover alternative buyers and APMC/market contacts.
- Helps farmers compare selling options using estimated net return rather than headline price alone.
- Reduces friction in asking market questions through regional-language voice interaction.
- Gives buyers a directory and enquiry pathway to connect with farmers.
- Encourages more transparent decisions by showing source, reporting date and known limitations.

12. Risks, limitations and trust principles

- A reported wholesale benchmark is not necessarily the price a specific farmer will receive from a buyer.
- Buyer offers may expire or change; farmers should confirm current price, quantity, quality requirements, pickup and payment terms before transporting produce.
- Forecasts are baseline estimates and must not be presented as guaranteed future prices.
- Missing or stale source data should be labelled clearly rather than silently treated as current.
- GPS access should be permission-based; manual village/taluk/district selection should remain available.
- Precise farm location should only be shared when necessary and with the farmer's consent.
- Verification labels should reflect actual verification status; unverified ratings or listings must not be presented as confirmed facts.
- Net-return calculations depend on the costs and assumptions entered; actual costs can vary.

13. Recommended live demo sequence

7. Open the dashboard and point out location and Kannada/English language options.
8. Use the voice assistant to ask for a crop price; show the transcript and spoken response.
9. Open the market directory and show the available APMC contact information and source details.
10. Open the interactive map and explain the mandi and buyer markers.
11. Search for buyers for a selected crop and show the available contact/enquiry path.
12. Enter a vendor quote and quantity in the comparison/net-return calculator.
13. If data is sufficient, demonstrate the baseline prediction and clearly state its limitations.

14. Three-slide presentation narrative

Slide 1 — Problem & Solution	Slide 2 — How It Works	Slide 3 — Built & Impact

Farmers face fragmented prices, limited buyer discovery and unclear net proceeds. KrishiSetu brings market benchmarks, voice assistance, buyer discovery and earnings estimates together.	**Ask/search → intent routing and data retrieval → compare market and buyer options → estimate net return → contact the buyer. Highlight voice, market directory, map, weather and baseline forecast.**	**Show actual stack and prototype metrics: 16,307 price records, 10 APMC contacts, 41 mandi locations, 200+ commodities, 5 source adapters, 14 intents. Mention forecast/data coverage limitations honestly.**

15. Short project description for forms

KrishiSetu is a farmer-focused agricultural market discovery platform that combines government crop-price observations, APMC/market contacts, buyer discovery, interactive maps, Kannada/English voice assistance and a net-return calculator. Built with Next.js, TypeScript, Express and MongoDB Atlas, it uses Gnani Prisma v2.5 speech-to-text and Timbre v2.5 text-to-speech to make market queries more accessible. Farmers can inspect available market benchmarks, discover buyers, compare selling options and estimate earnings after selected costs. The prototype includes 16,307 price records, 41 mandi locations and a baseline price-prediction service, with transparent source attribution and clear limitations as coverage and historical data continue to improve.

16. Source basis and accuracy note

This document is based on the supplied files: “Prototype Overview & Presentation Guide,” “Feature: Real-Time Market Price Intelligence, Fair Price Advisor & Local Vendor Discovery,” and the hackathon details draft. Implementation claims and metrics are taken from the prototype overview. The broader feature specification includes planned or desired capabilities; where these are not confirmed as implemented in the prototype overview, they are treated as proposed capabilities or future improvements rather than completed functionality.